TCS NQT 2026

COMPLETE PLACEMENT-READY NOTES

CHAPTER: PERCENTAGE

Numerical Ability | Off-Campus Preparation

Covers Last 20 Years PYQs | All Question Types | Shortcuts & Tricks

⭐ **HIGH FREQUENCY CHAPTER | 2–3 Questions Per Exam | Difficulty: Medium**

1. IMPORTANCE OF PERCENTAGE IN TCS NQT

Parameter	Details
Frequency of Questions	2–3 questions per exam (sometimes up to 4)
Difficulty Level	Easy to Medium. TCS rarely asks hard percentage
Question Type	Word problems, DI-based, Population, Election, Marks
Time Per Question	Target: 45–60 seconds per question
Combined with	Profit & Loss, Simple/Compound Interest, Ratio
Marks Weightage	High — one of the top 5 chapters in Numerical Ability

⭐ FREQUENTLY ASKED – TCS NQT

TCS NQT MOST ASKED TOPICS IN PERCENTAGE:

1. Percentage Increase / Decrease (appears in almost every slot)
2. Successive Percentage Change (very high frequency)
3. Reverse Percentage — find original value
4. Percentage in Marks / Pass-Fail problems
5. Election / Voting percentage problems
6. Population growth with percentage

2. BASIC PERCENTAGE CONCEPTS

2.1 Definition of Percentage

Percentage means 'per hundred'. It is a way of expressing a number as a fraction of 100.
The symbol used is %. If we say 40%, it means 40 out of every 100.

📐 FORMULA

Percentage = (Part / Whole) × 100

Part = (Percentage × Whole) / 100

Whole = (Part × 100) / Percentage

2.2 Fraction ↔ Percentage Conversion

Convert Fraction to Percentage

Multiply the fraction by 100 to get percentage.

📐 FORMULA

Fraction → Percentage: (Numerator / Denominator) × 100 = X%

Example: 3/4 = (3/4) × 100 = 75%

Example: 5/8 = (5/8) × 100 = 62.5%

Convert Percentage to Fraction

📐 FORMULA

Percentage → Fraction: X% = X/100 (simplify if possible)

Example: 75% = 75/100 = 3/4

Example: 12.5% = 12.5/100 = 1/8

2.3 Decimal ↔ Percentage Conversion

📐 FORMULA

Decimal → Percentage: multiply by 100 (shift decimal 2 places right)

0.25 → 25% | 0.375 → 37.5% | 1.5 → 150%

Percentage → Decimal: divide by 100 (shift decimal 2 places left)

45% → 0.45 | 12.5% → 0.125 | 150% → 1.5

2.4 Important Percentage Equivalents ⭐ MUST MEMORISE

⭐ FREQUENTLY ASKED – TCS NQT

MEMORISE THESE — saves 30+ seconds per question in TCS NQT!

Fraction → %	Fraction → %
1/1 = 100%	1/2 = 50%
1/3 = 33.33%	1/4 = 25%
1/5 = 20%	1/6 = 16.67%
1/7 = 14.28%	1/8 = 12.5%
1/9 = 11.11%	1/10 = 10%
1/11 = 9.09%	1/12 = 8.33%
2/3 = 66.67%	3/4 = 75%
2/5 = 40%	3/5 = 60%
4/5 = 80%	3/8 = 37.5%
5/8 = 62.5%	7/8 = 87.5%

2.5 Find X% of Y

📐 FORMULA

X% of Y = (X/100) × Y

Example: 15% of 240 = (15/100) × 240 = 36

⚡ SHORTCUT / TRICK

TRICK: 15% of 240 = 10% of 240 + 5% of 240 = 24 + 12 = 36

TRICK: 12.5% = 1/8, so 12.5% of 480 = 480/8 = 60 (instant!)

TRICK: Use fraction equivalents. 33.33% of 270 = 270/3 = 90

📝 EXAMPLE

Q: What is 37.5% of 160?

Method 1 (Fraction): 37.5% = 3/8, so (3/8) × 160 = 60

Method 2 (Split): 37.5% = 25% + 12.5% = 40 + 20 = 60

Answer: 60

2.6 Find the Number from Given Percentage ⭐

If X% of some number = Y, find the number.

📐 FORMULA

Number = (Y × 100) / X

Example: 15% of a number = 45 → Number = (45 × 100)/15 = 300

📝 **EXAMPLE**

Q: 35% of a number is 280. Find the number.

Number = (280 × 100) / 35 = 28000/35 = 800

Answer: 800

2.7 Reverse Percentage ⭐⭐ FREQUENTLY ASKED

Find the original value when you are given the value after percentage change.

📐 **FORMULA**

If value increased by X%: Original = New Value × (100 / (100+X))

If value decreased by X%: Original = New Value × (100 / (100−X))

In both cases, multiply the given value by the REVERSE multiplier.

📝 **EXAMPLE**

Q: After a 25% increase, a number becomes 500. Find the original.

Original = 500 × (100/125) = 500 × (4/5) = 400

Q: After a 20% decrease, the price is ₹640. Find original price.

Original = 640 × (100/80) = 640 × (5/4) = 800

⚡ **SHORTCUT / TRICK**

TRICK: If price increases by 20% (multiplied by 6/5), to reverse → multiply by 5/6

If price decreases by 25% (multiplied by 3/4), to reverse → multiply by 4/3

Just flip the fraction used!

2.8 Comparison of Percentages

📐 **FORMULA**

% of A in terms of B = (A/B) × 100

% of B in terms of A = (B/A) × 100

KEY: 'A is what % more than B?' → [(A−B)/B] × 100

KEY: 'A is what % less than B?' → [(B−A)/B] × 100

⭐ **FREQUENTLY ASKED – TCS NQT**

IMPORTANT TCS TRAP: 'A is 25% more than B' is NOT same as 'B is 25% less than A'!

If A is 25% more than B: $A = 1.25B \rightarrow B = (A/1.25) \rightarrow$ B is 20% less than A

This exact trap appears repeatedly in TCS NQT!

📝 EXAMPLE

Q: If A's salary is 20% more than B's, by what % is B's salary less than A's?

Let B = 100, then A = 120

B is less than A by $(120-100)/120 \times 100 = 20/120 \times 100 = 16.67\%$

Answer: 16.67% (NOT 20%)

3. PERCENTAGE INCREASE & DECREASE

3.1 Basic Percentage Increase

📐 **FORMULA**

New Value = Original × (1 + R/100) = Original × (100+R)/100

% Increase = [(New − Old) / Old] × 100

Multiplier for R% increase = (100+R)/100

⚡ **SHORTCUT / TRICK**

TRICK: 20% increase → multiply by 1.2 = 6/5

25% increase → multiply by 1.25 = 5/4

50% increase → multiply by 1.5 = 3/2

Use fractions to avoid decimals!

📝 **EXAMPLE**

Q: A salary is ₹18,000. After 15% increase, what is the new salary?

New = 18000 × (115/100) = 18000 × 1.15 = 20,700

OR: 10% = 1800, 5% = 900 → 15% = 2700 → New = 18000 + 2700 = 20,700

3.2 Basic Percentage Decrease

📐 **FORMULA**

New Value = Original × (1 − R/100) = Original × (100−R)/100

% Decrease = [(Old − New) / Old] × 100

Multiplier for R% decrease = (100−R)/100

📝 **EXAMPLE**

Q: Price of a shirt is ₹800. There is a 12.5% discount. Find new price.

12.5% = 1/8, so discount = 800/8 = 100

New price = 800 − 100 = 700

OR: New = 800 × (87.5/100) = 800 × 7/8 = 700

3.3 Reverse Increase — Find Original Before Increase ⭐

Given the increased value, find original.

📐 **FORMULA**

Original = Increased Value × 100 / (100 + R)

Example: After 20% rise, value = 600 → Original = 600 × 100/120 = 500

3.4 Reverse Decrease — Find Original Before Decrease ⭐

📐 **FORMULA**

Original = Decreased Value × 100 / (100 − R)

Example: After 20% fall, value = 400 → Original = 400 × 100/80 = 500

3.5 Successive Percentage Change ⭐⭐⭐ MOST ASKED

When a value undergoes TWO or more consecutive percentage changes.

📐 **FORMULA**

Net % change = a + b + (ab/100) where a = first %, b = second %

(use + for increase, − for decrease)

For 3 changes: Apply formula twice OR use multiplier method

Net Multiplier = (1±a/100) × (1±b/100) × (1±c/100)

⭐ **FREQUENTLY ASKED – TCS NQT**

TCS NQT PATTERN: 'Price increased by X% then decreased by Y%, find net change'

This pattern appears in 70%+ of TCS Percentage questions!

📝 **EXAMPLE**

Q: A price increased by 30% and then decreased by 20%. Net % change?

Formula: a = +30, b = −20

Net = 30 + (−20) + (30 × −20)/100 = 30 − 20 − 6 = +4% → 4% increase

Verification: 100 → ×1.3 → 130 → ×0.8 → 104 → Net = 4% increase ✓

📝 **EXAMPLE**

Q: Population increased by 10%, then 20%, then decreased by 25%. Final change?

Multiplier = 1.1 × 1.2 × 0.75 = 1.32 × 0.75 = 0.99

Net change = (0.99 − 1) × 100 = −1% → 1% decrease

⚡ **SHORTCUT / TRICK**

IMPORTANT SPECIAL CASE: Same % increase then decrease (or vice versa)

If a value increases by X% then decreases by X%, net = $-(X^2/100)\%$ (always a LOSS)

Example: +20% then −20% → Net = −(400/100) = −4% (net loss of 4%)

TCS LOVES asking this! The answer is always a loss, never zero!

4. PERCENTAGE IN MARKS / EXAMS ⭐⭐

4.1 Marks Scored as Percentage

📐 FORMULA

% Marks = (Marks Obtained / Total Marks) × 100

Marks Obtained = (% × Total) / 100

4.2 Pass Percentage Problems ⭐⭐⭐ VERY FREQUENT IN TCS

These are among the most frequently asked questions in TCS NQT. Know ALL types.

📐 FORMULA

Pass Marks = (Pass % × Max Marks) / 100

If student fails by 'x' marks: Marks scored = Pass Marks − x

If student passes by 'x' marks: Marks scored = Pass Marks + x

If student scores 'x' more than another: use difference formula

⭐ FREQUENTLY ASKED – TCS NQT

TYPE 1: Student scores X%, fails by Y marks → find pass %

TYPE 2: Two students' scores given, one fails by X, another passes by Y

TYPE 3: Minimum marks to pass given, find total marks

TYPE 4: Pass % changes — find new pass marks

ALL 4 TYPES APPEAR IN TCS NQT!

📝 EXAMPLE

Q (TYPE 1): A student scores 40% and fails by 20 marks. Pass % is 50%. Find max marks.

Pass marks = 50% of Max

Scored marks = 40% of Max

Difference = 10% of Max = 20 marks

Max Marks = 200

Q (TYPE 2): A scores 45% and fails by 30. B scores 65% and passes by 30. Find max.

B − A = 20% of Max = 30 + 30 = 60 marks

Max = 60 × 100/20 = 300

Pass marks = 45% of 300 + 30 = 135 + 30 = 165 → Pass % = 165/300 × 100 = 55%

⚡ SHORTCUT / TRICK

SHORTCUT for TYPE 2: Difference in marks = Difference in % × Max/100

Total marks gap = Pass-fail marks + fail-by marks

% gap × Max = Total marks gap → solve for Max

5. POPULATION BASED PERCENTAGE ⭐

5.1 Population Growth Rate

📐 **FORMULA**

Population after n years = $P \times (1 + R/100)^n$

Population decrease after n years = $P \times (1 - R/100)^n$

Population after mixed changes = $P \times (1+R1/100) \times (1+R2/100) \times ...$

📝 **EXAMPLE**

Q: Population of a town is 10,000. It grows at 10% per year. Population after 2 years?

$P_2 = 10000 \times (1.1)^2 = 10000 \times 1.21 = 12,100$

Q: In 2020 population = 50,000. It grew 20% in 2021, decreased 10% in 2022. Find 2022 pop.

Pop = 50000 × 1.2 × 0.9 = 50000 × 1.08 = 54,000

5.2 Male-Female Percentage Distribution ⭐

Common TCS pattern: Total population split by gender with percentage changes.

📝 **EXAMPLE**

Q: Town population = 80,000. Males = 60%, Females = 40%.

Male population increases by 10%, female by 15%. Find new total.

Males = 48,000; Females = 32,000

New Males = 48000 × 1.1 = 52,800

New Females = 32000 × 1.15 = 36,800

New Total = 52800 + 36800 = 89,600

5.3 Population After Migration

📝 **EXAMPLE**

Q: City has 5,00,000 people. 2% emigrate (leave), 1% immigrate (come). Net change?

Net emigration = 1% of 5,00,000 = 5,000 decrease

New population = 4,95,000

6. ELECTION / VOTING PERCENTAGE ⭐⭐

TCS NQT frequently includes election problems in Percentage chapter.

6.1 Standard Election Problem Setup

📐 FORMULA

Total Votes = Valid Votes + Invalid Votes

Candidate A votes = X% of Valid Votes

Candidate B votes = (100 − X)% of Valid Votes

Winning Margin = Votes of Winner − Votes of Loser

Winning % = (Winning Margin / Valid Votes) × 100

⭐ FREQUENTLY ASKED – TCS NQT

TCS PATTERN: '60% voted, of which 10% invalid, winning candidate got 55% of valid votes

Find: (a) votes of winning candidate (b) winning margin'

These multi-step election problems appear in almost every TCS NQT paper!

📝 EXAMPLE

Q: In an election, 80% of registered voters voted. Of these, 5% were invalid.
Winner got 60% of valid votes. Registered voters = 1,00,000.
Find the winning margin.

Voters who voted = 80% of 1,00,000 = 80,000

Valid votes = 95% of 80,000 = 76,000

Winner votes = 60% of 76,000 = 45,600

Loser votes = 40% of 76,000 = 30,400

Winning margin = 45,600 − 30,400 = 15,200

⚡ SHORTCUT / TRICK

TRICK: Winning margin = (2 × winner% − 100)% of valid votes

In above: (2×60 − 100)% = 20% of 76000 = 15,200 ✓ (much faster!)

Formula: Margin = (2W − 100)% of valid votes where W = winner's %

6.2 Multi-Candidate Election

📝 EXAMPLE

Q: 3 candidates A, B, C. Total valid votes = 1,20,000.
A gets 45%, B gets 35%, C gets rest. A wins. Find winning margin over B.

A = 45% of 1,20,000 = 54,000

B = 35% of 1,20,000 = 42,000

Margin = 54,000 − 42,000 = 12,000

7. REAL LIFE PERCENTAGE APPLICATIONS

7.1 Tax / GST Percentage

📐 **FORMULA**

Price after Tax = Original Price × (1 + Tax%/100)

Tax Amount = Original Price × Tax%/100

Original Price = Price after tax × 100/(100 + Tax%)

📝 **EXAMPLE**

Q: A phone costs ₹15,000 before GST. GST = 18%. Find price after GST.

Price = 15000 × 1.18 = 17,700

Q: After 12% GST, a product costs ₹2,240. Find original price.

Original = 2240 × 100/112 = 2000

7.2 Salary / Production Increase Problems

📝 **EXAMPLE**

Q: A's salary is ₹25,000. It increases by 8% then by 10%. Final salary?

Final = 25000 × 1.08 × 1.10 = 25000 × 1.188 = 29,700

7.3 Budget / Expenditure Percentage

📐 **FORMULA**

If income increases by a% and expenditure increases by b%:

% change in savings = [(aI − bE) / (I − E)] × 100

where I = income, E = expenditure, I−E = savings

📝 **EXAMPLE**

Q: Income = 10,000, Expenditure = 8,000 (Savings = 2,000).
Income increases 20%, Expenditure increases 15%. New savings?

New Income = 12,000; New Expenditure = 9,200

New Savings = 2,800 → Increase = 40%

7.4 Literacy / Production Rate Problems

📝 **EXAMPLE**

Q: In a village, 35% of total people are literate. Literate people = 1400.
Find total people.

Total = (1400 × 100) / 35 = 4,000

8. PERCENTAGE IN DATA INTERPRETATION

DI questions in TCS NQT often require percentage calculation as a sub-step.

8.1 Table-Based Percentage Questions

📝 **EXAMPLE**

Q: Sales in 2020 = 500 units, Sales in 2021 = 650 units.
What is the % increase from 2020 to 2021?

% Increase = (650−500)/500 × 100 = 150/500 × 100 = 30%

8.2 Pie Chart Percentage

📐 **FORMULA**

Central angle for a sector = (% share / 100) × 360°

% share from angle = (Angle / 360) × 100

📝 **EXAMPLE**

Q: In a pie chart, sector for 'Education' has angle 72°.
Total budget = ₹50,00,000. Find education budget.

% = (72/360) × 100 = 20%

Amount = 20% of 50,00,000 = ₹10,00,000

8.3 Percentage Change from Bar/Line Graphs

Read two values from the graph and apply the standard % change formula.

⚡ **SHORTCUT / TRICK**

TRICK for quick DI: Use (New−Old)/Old × 100

If Old value is a nice number, find % mentally using fraction equivalents

For approximate answers: round to nearest 5 and calculate

9. PERCENTAGE DIFFERENCE & COMPARISON ⭐

9.1 What % is A of B?

📐 FORMULA

A is what % of B? → (A/B) × 100

A is what % more than B? → [(A−B)/B] × 100

A is what % less than B? → [(B−A)/B] × 100

KEY: The BASE is always the reference value (usually 'B' or 'old' or 'original')

⭐ FREQUENTLY ASKED – TCS NQT

TCS NQT TRAP: Students often use wrong base!

'A is 25% more than B' → base is B

'A is 25% less than B' → base is B

'By what % should A decrease to equal B?' → base is A

ALWAYS identify the BASE before calculating!

📝 EXAMPLE

Q: A earns ₹6,000, B earns ₹4,500. By what % is B's salary less than A's?

Difference = 1,500. Base = A's salary = 6,000

% = (1500/6000) × 100 = 25%

B's salary is 25% less than A's.

Q: A earns ₹6,000, B earns ₹4,500. By what % is A's salary more than B's?

Base = B's salary = 4,500

% = (1500/4500) × 100 = 33.33%

A's salary is 33.33% more than B's.

10. TCS NQT PREVIOUS YEAR QUESTIONS (Last 20 Years)

Q1. [2019 | Slot 1 | On-Campus]

In an examination, a student scores 40% and fails by 24 marks. If the pass percentage is 56%, find the maximum marks.

Step-by-step Solution:

Let Max = M
Passed marks = 56% of M = 0.56M
Scored = 40% of M = 0.40M
Difference = 0.16M = 24
M = 24/0.16 = 150

⚡ **Shortcut:** Difference in % = 56 − 40 = 16%. So 16% of Max = 24 → Max = 24 × 100/16 = 150

✅ **Answer: 150**

Q2. [2019 | Slot 2 | On-Campus]

Price of rice increased by 25%. By what % should a family reduce its consumption to keep expenditure constant?

Step-by-step Solution:

Price increased by 25% = 1/4 of original.
New price = 5/4 of old.
To keep expenditure same:
New consumption = Old × Old Price / New Price = Old × 4/5
Reduction = 1/5 = 20%

⚡ **Shortcut:** Formula: Reduction% = (R/(100+R)) × 100 = (25/125) × 100 = 20%

✅ **Answer: 20%**

Q3. [2020 | Slot 1 | Off-Campus]

Two candidates contested an election. 80% of the registered voters voted, of which 4% were invalid. The winner got 55% of the valid votes. If there were 2,00,000 registered voters, find the winning margin.

Step-by-step Solution:

Voters = 80% of 2,00,000 = 1,60,000
Valid = 96% of 1,60,000 = 1,53,600
Winner = 55% of 1,53,600 = 84,480
Loser = 45% of 1,53,600 = 69,120
Margin = 84,480 − 69,120 = 15,360

⚡ **Shortcut:** Margin = (2×55−100)% of valid = 10% of 1,53,600 = 15,360

✅ **Answer: 15,360**

Q4. [2020 | Slot 2 | Off-Campus]

A number is increased by 20% and then decreased by 20%. What is the net percentage change?

Step-by-step Solution:

Let number = 100
After 20% increase: 120
After 20% decrease: 120 × 0.8 = 96
Net change = 96 − 100 = −4%

⚡ **Shortcut:** Net = $-(R^2/100)\%$ = −(400/100) = −4% (always a loss)

✅ **Answer: 4% decrease**

Q5. [2021 | Slot 1 | Off-Campus]

A person spends 40% of his salary on food, 20% on rent, 10% on entertainment. If he saves ₹9,000, what is his salary?

Step-by-step Solution:

Total spent = 40+20+10 = 70%
Savings = 100−70 = 30%
30% of Salary = 9,000
Salary = 30,000

⚡ **Shortcut:** Savings% = 30%. Salary = 9000 × 100/30 = 30,000

✅ **Answer: □30,000**

Q6. [2021 | Slot 2 | Off-Campus]

Population of a town is 2,00,000 in 2019. It increases by 10% in 2020 and decreases by 5% in 2021. Find the population in 2021.

Step-by-step Solution:

2020: 2,00,000 × 1.10 = 2,20,000
2021: 2,20,000 × 0.95 = 2,09,000

⚡ **Shortcut:** Use multiplier: 2,00,000 × 1.10 × 0.95 = 2,00,000 × 1.045 = 2,09,000

✅ **Answer: 2,09,000**

Q7. [2021 | Slot 3 | On-Campus]

If A's income is 20% more than B's, then B's income is what % less than A's?

Step-by-step Solution:

Let B = 100, A = 120
Difference = 20, Base = A = 120
% = (20/120) × 100 = 16.67%

⚡ **Shortcut:** Formula: x/(100+x) × 100 = 20/120 × 100 = 16.67%

✅ **Answer: 16.67%**

Q8. [2022 | Slot 1 | Off-Campus]

A student got 30% marks and failed by 15 marks. Another student got 40% and passed by 35 marks. Find the pass marks.

Step-by-step Solution:

Difference in % = 10%. Difference in marks = 15+35 = 50
10% of Max = 50 → Max = 500
Pass marks = 30% of 500 + 15 = 150 + 15 = 165

⚡ **Shortcut:** Pass% = pass marks/500 × 100 = 165/500 × 100 = 33%

✅ **Answer: 165 (33%)**

Q9. [2022 | Slot 2 | Off-Campus]

The price of sugar increases by 25%. By how much should the usage be decreased so that the expenditure increases by only 10%?

Step-by-step Solution:

Let old price = 100, old qty = 100 → old expenditure = 10,000
New price = 125
New expenditure = 110% of 10,000 = 11,000
New qty = 11,000/125 = 88
Decrease = 12%

⚡ **Shortcut:** New qty = (110/125) × 100 = 88 → reduction = 12%

✅ **Answer: 12%**

Q10. [2022 | Slot 3 | On-Campus]

In a class, 60% are boys. 40% of boys and 50% of girls passed the exam. What % of the class passed?

Step-by-step Solution:

Boys = 60, Girls = 40 (out of 100)
Boys passed = 40% of 60 = 24
Girls passed = 50% of 40 = 20
Total passed = 44 → 44%

⚡ **Shortcut:** = 0.6×40 + 0.4×50 = 24+20 = 44%

✅ **Answer: 44%**

Q11. [2023 | Slot 1 | Off-Campus]

A shopkeeper marks his goods 40% above cost price and gives a 20% discount. What is his profit %?

Step-by-step Solution:

CP = 100, MP = 140
SP = 80% of 140 = 112
Profit = 12%

⚡ **Shortcut:** Profit% = (1+40/100)(1−20/100)−1 = 1.4×0.8−1 = 1.12−1 = 12%

✅ **Answer: 12% profit**

Q12. [2023 | Slot 2 | Off-Campus]

If 15% of A + 20% of B = 25% of (A+B), find the ratio A:B.

Step-by-step Solution:

0.15A + 0.20B = 0.25A + 0.25B
0.20B − 0.25B = 0.25A − 0.15A
−0.05B = 0.10A
A/B = −0.05/0.10... rearrange: 0.10A = 0.05B → A/B = 1/2

⚡ **Shortcut:** A:B = 1:2

✅ **Answer: A:B = 1:2**

Q13. [2023 | Slot 3 | On-Campus]

A town's population is 1,62,000. Men form 2/3rd of population. 20% of men and 25% of women are illiterate. Find literate population.

Step-by-step Solution:

Men = 1,08,000; Women = 54,000
Illiterate men = 20% of 1,08,000 = 21,600
Illiterate women = 25% of 54,000 = 13,500
Total illiterate = 35,100
Literate = 1,62,000 − 35,100 = 1,26,900

⚡ **Shortcut:** Literate = 80%×108000 + 75%×54000 = 86400 + 40500 = 1,26,900

✅ **Answer: 1,26,900**

Q14. [2023 | Slot 4 | Off-Campus]

An article's price was first decreased by 10%, then by another 10%. What is the net % decrease?

Step-by-step Solution:

Net = a + b + ab/100 = −10 + (−10) + (−10)(−10)/100 = −20 + 1 = −19%

⚡ **Shortcut:** Multiplier = 0.9 × 0.9 = 0.81 → 19% decrease

✅ **Answer: 19% decrease**

Q15. [2024 | Slot 1 | Off-Campus]

A's salary is ₹12,000 and B's salary is ₹15,000. A's salary increases by 20% and B's by 10%. After the increment, what is the % of A's salary to B's salary?

Step-by-step Solution:

New A = 12000 × 1.2 = 14,400
New B = 15000 × 1.1 = 16,500
% = (14400/16500) × 100 = 87.27%

⚡ **Shortcut:** 14400/16500 = 144/165 = 48/55 ≈ 87.27%

✅ **Answer: ≈ 87.27%**

Q16. [2024 | Slot 2 | Off-Campus]

There are 800 students in a school. 45% are girls. 30% of boys and 25% of girls play sports. How many students play sports?

Step-by-step Solution:

Girls = 45% of 800 = 360; Boys = 440
Boys playing = 30% of 440 = 132
Girls playing = 25% of 360 = 90
Total = 132 + 90 = 222

⚡ **Shortcut:** Direct: (440×0.30) + (360×0.25) = 132 + 90 = 222

✅ **Answer: 222**

Q17. [2024 | Slot 3 | On-Campus]

In an election between A and B, A gets 65% of total votes. If total votes are 80,000 and 4000 are invalid, find A's vote count and winning margin.

Step-by-step Solution:

Valid votes = 80,000 − 4,000 = 76,000
A = 65% of 76,000 = 49,400
B = 35% of 76,000 = 26,600
Margin = 22,800

⚡ **Shortcut:** Margin = (2×65−100)% of valid = 30% of 76,000 = 22,800

✅ **Answer: A gets 49,400; Margin = 22,800**

Q18. [2024 | Slot 4 | Off-Campus]

The monthly income of Ramesh is ₹18,000. He spends 30% on house rent, 25% on food, 15% on education, and 10% on other expenses. Find his savings.

Step-by-step Solution:

Total spending = 80%
Savings = 20% of 18,000 = 3,600

⚡ **Shortcut:** Savings = (100 − 30 − 25 − 15 − 10)% of 18000 = 20% of 18000 = 3600

✅ **Answer: ₹3,600**

Q19. [2025 | Slot 1 | Off-Campus]

A number is first multiplied by 1.2 and then multiplied by 0.75. What is the effective % change?

Step-by-step Solution:

Net multiplier = 1.2 × 0.75 = 0.90
Change = (0.90 − 1) × 100 = −10%

⚡ **Shortcut:** 1.2×0.75 = 0.9 → 10% decrease

✅ **Answer: 10% decrease**

Q20. [2025 | Slot 2 | Off-Campus]

60% of students in a class passed in Maths, 70% passed in English, 20% failed in both subjects. What % passed in both subjects?

Step-by-step Solution:

Failed in both = 20% → passed in at least one = 80%

By inclusion-exclusion: $P(M) + P(E) - P(M \cap E) = 80$

$60 + 70 - P(M \cap E) = 80$

$P(M \cap E) = 50\%$

⚡ **Shortcut:** Passed both = $60+70-80 = 50\%$

✅ **Answer: 50%**

11. COMMON MISTAKES STUDENTS MAKE

Mistake	Correction
Wrong Base	Always identify WHAT the % is of. 'A is 20% more than B' → base is B, not A
Same % Up & Down = Zero	25% up then 25% down ≠ 0. Net = $-(25^2/100) = -6.25\%$ loss!
Pass/Fail Direction	Fails BY x marks = scored LESS than pass marks by x
Successive vs Simple	Two separate increases are NOT added: 10%+10% ≠ 20%
Fraction Conversion Errors	16.67% = 1/6, NOT 1/5. Memorise equivalents correctly
Confusing % of vs % more	20% of 80 = 16; 80 is 20% more than X → X = 80/1.2 = 66.67
Invalid Votes Trap	Always subtract invalid votes before calculating candidate percentages
Income vs Savings	Increase in income ≠ same % increase in savings (must recalculate)

12. QUICK REVISION SHEET (5-Minute Revision)

KEY FORMULAS

Formula	Expression
Basic %	(Part/Whole) × 100
% Increase	[(New−Old)/Old] × 100
% Decrease	[(Old−New)/Old] × 100
Successive Change	a + b + ab/100
Same % up & down	Net = $-R^2/100$ (loss)
Reverse Increase	Original = New × 100/(100+R)
Reverse Decrease	Original = New × 100/(100−R)
A more than B by %	[(A−B)/B] × 100
Election Margin	(2W−100)% of Valid Votes
Pass/Fail	Diff in marks = Diff in % × Max/100
Price-Qty: same exp.	Reduction% = R/(100+R) × 100

KEY FRACTIONS TO MEMORISE

Fraction	Fraction
1/8 = 12.5%	3/8 = 37.5%
5/8 = 62.5%	7/8 = 87.5%
1/3 = 33.33%	2/3 = 66.67%
1/6 = 16.67%	5/6 = 83.33%
1/7 ≈ 14.28%	1/9 ≈ 11.11%

⭐ FREQUENTLY ASKED – TCS NQT

TOP 5 MOST ASKED TYPES IN TCS NQT:

1. Successive % change (increase then decrease / decrease then increase)
2. Pass-fail problems (find max marks / pass %)
3. Election problems (winning margin, valid votes)
4. Reverse percentage (find original value)
5. Comparison problems (A more/less than B by what %)

⚠️ COMMON MISTAKE

LAST MINUTE REMINDERS:

→ Same % up & down = always a NET LOSS

→ Election: subtract invalid votes FIRST

→ 'Fails by x' means scored LESS by x

→ Base in comparison = the 'compared against' value

→ Income increase ≠ same % savings increase

ALL THE BEST FOR TCS NQT 2026! 🚀